

A LITERATURE-IMPLIED PARTIAL ANSWER TO A WEAK-HILBERT JOHNSON–LINDENSTRAUSS QUESTION

AGENT_LANE_00

SOURCE QUESTION

Johnson and Naor [1] end their paper with remarks and open problems about Banach spaces satisfying the Johnson–Lindenstrauss lemma. In the first item they emphasize weak Hilbert spaces as the natural class around their positive example $T^{(2)}$. They record that it was not known whether every weak Hilbert space is log-Hilbertian, or even h -Hilbertian for some $h(n) = o(n)$, while weak Hilbert spaces with an unconditional basis were known, by Nielsen–Tomczak-Jaegermann, to satisfy a stronger finite-dimensional Hilbertian decomposition estimate.

LATER LITERATURE

Suarez de la Fuente [2] revisits exactly this weak-Hilbert/J-L direction. The introduction recalls the Johnson–Naor definition of the J-L lemma for Banach spaces, notes that the Johnson–Naor result extends to weak Hilbert spaces with an unconditional basis, and says that the arbitrary weak Hilbert case was still not known.

The same paper then considers the twisted Hilbert space

$$Z(T^2) = d(T^2, (T^2)^*)_{1/2}.$$

It records that $Z(T^2)$ is a weak Hilbert space, that it fails to have an unconditional basis by Kalton’s theorem on twisted Hilbert spaces, and proves the theorem:

$Z(T^2)$ satisfies the Johnson–Lindenstrauss lemma.

The proof is by showing that $Z(T^2)$ is log-Hilbertian and then invoking Johnson–Naor’s result that log-Hilbertian spaces satisfy the J-L lemma.

Claim. *There exists a weak Hilbert space with no unconditional basis that satisfies the Johnson–Lindenstrauss lemma. One such space is $Z(T^2)$.*

Literature implication. By Suarez de la Fuente [2], $Z(T^2)$ is a weak Hilbert twisted Hilbert space and has no unconditional basis. Theorem log of [2] proves that $Z(T^2)$ satisfies the J-L lemma. Hence the class of weak Hilbert spaces satisfying the J-L lemma strictly extends beyond the unconditional-basis subclass mentioned in the original Johnson–Naor discussion. $\square$

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Scope. *This is only a partial answer. It does not prove that every weak Hilbert space is log-Hilbertian, h -Hilbertian with $h(n) = o(n)$, or J-L. It only supplies a concrete positive example without an unconditional basis, thereby settling that subcase/variant of the Johnson–Naor weak-Hilbert direction.*

REFERENCES

- [1] W. B. Johnson and A. Naor, *The Johnson–Lindenstrauss lemma almost characterizes Hilbert space, but not quite*, Discrete Comput. Geom. 43 (2010), no. 3, 542–553; arXiv:0807.1919.
- [2] J. Suarez de la Fuente, *A space with no unconditional basis that satisfies the Johnson–Lindenstrauss lemma*, arXiv:2501.13524.